

A MONTE CARLO STUDY OF SEQUENTIAL CONFIDENCE INTERVALS BASED ON RANK TESTS*

J.C. Geertsema**

The Florida State University, Tallahassee

1. Introduction

In Geertsema (1968) the author proposed two sequential procedures for the problem of finding a bounded length confidence interval for the point of symmetry of a symmetric distribution. These procedures are defined in the following way: If μ is the centre of symmetry, let I_n be a fixed sample size confidence interval for μ based on n observations such that $\lim_{n \rightarrow \infty} P(\mu \in I_n) = 1 - 2\alpha$, and denote by $\ell(I_n)$ the (random) length of the interval I_n . The stopping variable N of the sequential procedure is defined to be the first integer $n \geq n_0$ such that $\ell(I_n) \leq 2d$, where $2d$ is the bound on the length of the required confidence interval and n_0 is a positive integer. Then take as confidence interval I_N .

Chow and Robbins (1965) considered the case where $I_n = (\bar{X}_n \pm a_n \frac{S_n}{\sqrt{n}})$ where, if

$X_1, \dots, X_n$ denotes a random sample of size n from the underlying distribution F , $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$, $S_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2$, and a_n is a sequence of positive numbers such that $a_n \rightarrow K_\alpha$ and $\Phi(K_\alpha) = 1 - \alpha$ where Φ is the standard normal c.d.f. For this study, we choose $a_n = t_{n,\alpha}$ where $P(T > t_{n,\alpha}) = \alpha$ if T has a t -distribution with n degrees of freedom. This procedure will be denoted by M in the rest of the paper. (It is based on the sample mean.)

In Geertsema (1968) the following two cases are considered: (i) $I_n = (X_{n,b(n)}, X_{n,a(n)})$, where $X_{n,1} \leq X_{n,2} \leq \dots \leq X_{n,n}$ are the order statistics in a sample of size n and $a(n)$ and $b(n)$ are sequences of positive integers, defined by

$$b(n) = \max \left\{ 1, \left[\frac{n}{2} - K_\alpha \frac{\sqrt{n}}{2} \right] \right\}$$

$$a(n) = n - b(n) + 1$$

where $[x]$ is the largest integer less than or equal to x .

The interval I_n corresponds in a natural way to the sign test and therefore we will denote the procedure based on I_n by S .

(ii) $I_n = (W_{n,b'(n)}, W_{n,a'(n)})$, where $W_{n,1} \leq W_{n,2} \leq \dots \leq W_{n,n(n+1)/2}$ are the ordered $\frac{X_i + X_j}{2}$, $i \leq j$.

$a'(n)$ and $b'(n)$ are sequences of positive integers, defined by

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**Now at Potchefstroom University

$$b'(n) = \max \left\{ 1, [n(n+1)/4 - K_a(n(n+1)(2n+1)/24)^{1/2}] \right\}$$

$$a'(n) = n(n+1)/2 - b(n) + 1.$$

This procedure will be denoted by W because of the natural correspondence with the Wilcoxon one-sample test.

It is proved in Chow and Robbins (1965) and Geertsema (1968) that under suitable regularity conditions each procedure stops a.s. and $\lim_{d \rightarrow 0} N = \infty$ a.s., $\lim_{d \rightarrow 0} EN = \infty$, $\lim_{d \rightarrow 0} P(\mu \in I_N) = 1 - 2\alpha$ for each of the procedures, and that

$$\text{for } M : \lim_{d \rightarrow 0} ENd^2 = K_a^2 \sigma^2$$

$$\text{for } S : \lim_{d \rightarrow 0} ENd^2 = \frac{K_a^2}{4f^2(0)}$$

$$\text{for } W : \lim_{d \rightarrow 0} ENd^2 = \frac{K_a^2}{12(\int f^2(x)dx)^2}$$

where σ^2 is the variance of F and f the density of F.

It follows immediately from these results that if the asymptotic efficiency (as $d \rightarrow 0$) between any two of the three procedures is defined as the limiting ratio of expected sample sizes, then these asymptotic efficiencies coincide with the Pitman efficiencies between the corresponding tests (where M is thought of as corresponding to the t-test).

Important questions that arise are: how well do the above asymptotic results describe the behavior of the procedures as to coverage probability, expected sample size and "efficiency"?

2. Monte Carlo Studies

The above questions were investigated in a series of Monte Carlo studies for the following populations:

Standard normal with density $\phi(x)$

Contaminated normal with density $.9\phi(x) + \frac{1}{3}\phi(\frac{x}{3})$

Uniform on $(-1/2, 1/2)$

Double exponential with density $\frac{1}{2}\exp(-|x|)$

The results apply also to these populations under transformations of location and scale.

A sequence of 15 values of d was chosen in such a way that d^{-2} increases by a constant amount. Also α was chosen 0.025 (so that $1 - 2\alpha = 0.95$). The procedures were required to take at least six observations because $P(\mu \in I_n) \geq 0.95$ for the first time when $n \geq 6$ in the case of the procedures S and W.

From a pseudo-random number generator m samples from each of the above populations were generated by the CDC 6400 computer at the Florida State University, Tallahassee, where $m = 2000$ for the uniform distribution and $m = 1000$ in the other cases. All three procedures were applied to each sample sequence, starting with the largest value of d and continuing with the other values of d until the smallest was reached. For each value of d and each procedure the value of N was recorded and also whether the interval covered μ or not.

Because of the amount of computer time involved procedure W was not investigated for all the values of d.

The results are presented in tables 1 and 2. In table 1 the columns headed by M, S and W give, for each value of $2d$, an estimate $2\hat{a}$ of the probability $2a$ of not covering μ , where $2\hat{a} = f/m$ and f is the number of times the procedure did not cover μ . The columns headed "error" give a 95% error estimate $1.96 (2\hat{a}(1-2\hat{a})/m)^{1/2}$.

In table 2 an estimate $\hat{EN}$ for the expected sample size for each value of $2d$ is given in the columns marked M, S and W where $\hat{EN} = \sum n_i / n$ and n_i ($i=1, \dots, m$) denote the sample sizes required by a procedure. The columns marked "error" give a 95% error estimate, $1.96 \{(\sum n_i^2 / m - (\sum n_i / m)^2) / m\}^{1/2}$. The columns A_i give the asymptotic values for EN for procedure i ($i=M, S, W$). The quantities B_S and B_W are defined as follows:

$$B_S = \frac{1.96^2}{4[F(d) - 1/2]^2} \quad B_W = \frac{1.96^2}{3[G(d) - 1/2]^2}$$

where F is the c.d.f. of the underlying distribution and G is the c.d.f. of the average of two independent random variables with c.d.f. F . It is straight forward to calculate G for each of the populations considered.

A heuristic motivation for considering B_S is as follows:

Notice first that $\lim_{d \rightarrow 0} B_S d^2 = A_S d^2$.

Furthermore, consider the sequential procedure with stopping variable N , defined as the first integer n such that (if $\mu \neq 0$).

$$X_{n,b(n)} \geq -d.$$

Then $X_{N-1,b(N-1)} < -d$, so that

$$\begin{aligned} S_N(-d) &\geq S_{N-1}(-d) \\ &\geq b(N-1) \\ &\geq \frac{N-1}{2} - K_a \frac{(N-1)^{1/2}}{2} - 1 \end{aligned}$$

where $S_n(t) = \sum_{i=1}^n I(X_i, t)$ and $I(a, b) = 1$ if $a \leq b$ and $I(a, b) = 0$ otherwise.

Applying Wald's identity, one gets

$$\begin{aligned} F(-d) E N &\geq \frac{E(N-1)}{2} - \frac{K_a}{2} \cdot E(N-1)^{1/2} - 1 \\ &\geq \frac{E(N-1)}{2} - \frac{K_a}{2} \cdot [E(N-1)]^{1/2} - 1 \end{aligned}$$

by Jensen's inequality. This gives (using the symmetry of F)

$$E N \leq \frac{K_a^2}{4[F(d) - 1/2]^2} + O\left(\frac{1}{d}\right)$$

This sequential procedure is so closely related to the confidence interval procedure S , that one would intuitively expect that an upper bound for the expected sample size of procedure S would be of the form $C + B_S$ where C is approximately constant (as a function of d). This also motivates the consideration of B_W in the case of the W procedure.

TABLE 2
Estimates $\hat{EN}$ of the expected sample sizes

2d	M	Error	A_M	S	Error	A_S	B_S	W	Error	A_W	B_W
Normal distribution											
1.794	7.09	.10	4.77	13.89	.33	7.5	9.67	8.71	.14	5	8.09
1.269	10.33	.23	9.55	21.98	.53	15	17.10	12.90	.25	10	12.90
1.036	14.37	.34	14.32	29.83	.76	22.5	24.55	17.12	.35	15	17.80
.897	18.55	.42	19.10	37.45	.93	30	31.98	21.58	.42	20	22.74
.802	23.10	.49	23.87	44.84	1.09	37.5	39.57	26.22	.50	25	27.81
.732	27.66	.57	28.65	52.44	1.28	45	47.10	31.02	.56	30	33.00
.678	32.44	.63	33.42	59.69	1.43	52.5	54.54	35.83	.62	35	37.82
.634	37.48	.67	38.20	67.12	1.63	60	62.06	40.96	.69	40	42.70
.598	42.40	.71	42.97	74.08	1.79	67.5	69.56	46.05	.73	45	47.73
.567	47.30	.75	47.75	81.85	1.97	75	76.84				
.541	52.40	.77	52.52	89.04	2.13	82.5	84.20				
.518	57.25	.80	57.30	95.59	2.25	90	91.95				
.598	62.81	.82	62.07	102.81	2.39	97.5	99.39				
.479	67.10	.86	66.85	109.49	2.49	105	106.64				
.463	72.11	.88	71.62	116.22	2.60	112.5	114.21				
Contaminated normal distribution											
1.794	9.12	.26	8.59	16.17	.35	8.61	11.01	10.52	.21	6.26	9.75
1.268	15.27	.50	17.19	25.92	.61	17.22	21.67	16.15	.34	12.52	16.65
1.036	22.72	.72	25.78	36.03	.87	25.83	38.09	21.94	.45	18.77	21.99
.897	30.29	.88	34.38	44.56	1.05	34.44	36.64	28.02	.55	25.03	28.17
.802	38.47	1.04	42.97	53.57	1.25	43.05	45.30	34.01	.62	31.29	34.52
.732	46.68	1.15	51.57	62.29	1.45	51.66	53.97	40.09	.70	37.55	40.74
.678	54.87	1.27	60.16	70.46	1.62	60.27	62.56				
.634	63.37	1.37	68.75	79.92	1.83	68.88	71.13				
.598	72.79	1.43	77.35	89.17	2.04	77.49	79.81				
.567	81.63	1.49	85.94	97.11	2.22	86.10	88.12				
.541	90.27	1.56	94.54	105.93	2.37	94.71	97.20				
.518	98.59	1.64	103.13	113.74	2.55	103.32	105.53				
.498	106.74	1.73	111.73	123.12	2.79	111.93	113.96				
.479	115.50	1.77	120.32	131.90	3.03	120.54	122.62				
.463	124.03	1.82	128.92	140.94	3.18	129.15	131.07				
Uniform distribution											
.653	6.13	.02	3	11.86	.22	9	9	7.37	.06	3	6.62
.462	7.85	.11	6	21.65	.39	18	18	10.32	.11	6	10.15
.377	10.21	.12	9	31.18	.58	27	27	13.42	.16	9	13.68
.327	12.86	.16	12	40.22	.74	36	36	16.60	.19	12	17.12
.292	15.61	.19	15	49.16	.88	45	45	19.90	.22	15	20.59
.267	18.56	.22	18	57.79	1.03	54	54	23.34	.24	18	23.91
.247	21.46	.24	21	66.74	1.17	63	72	26.81	.26	21	27.32
.231	24.53	.26	24	75.62	1.32	72	81	30.22	.27	24	30.68
.218	27.65	.28	27	84.41	1.44	81	90	33.66	.28	27	33.94
.207	30.78	.29	30	93.63	1.58	90	99	37.18	.29	30	37.18
.197	33.83	.30	33	102.23	1.73	99	99	40.59	.29	33	40.60
.189	36.98	.30	36	110.62	1.85	108	108	53.97	.30	36	43.72
.181	40.06	.31	39	119.63	1.99	117	117	47.22	.31	39	47.25
.175	43.25	.32	42	128.52	2.10	126	126				
.169	46.35	.33	45	137.99	2.24	135	135				
Double exponential distribution											
2.024	8.47	.23	7.5	14.39	.33	3.75	9.48	10.20	.21	5	9.50
1.431	14.07	.44	15	20.42	.47	7.5	14.70	14.90	.34	10	14.72
1.169	20.15	.63	22.5	26.19	.62	11.25	19.63	19.68	.46	15	19.86
1.012	26.51	.79	30	31.52	.76	15	24.36	24.42	.57	20	25.01
.905	33.62	.95	37.5	36.59	.86	18.75	28.95	29.48	.67	25	30.15
.826	41.58	1.10	45	41.56	.96	22.5	33.56	34.46	.74	30	35.21
.765	48.77	1.18	52.5	46.40	1.07	26.25	37.95	39.62	.79	35	40.32
.716	56.23	1.28	60	51.08	1.15	30	42.42				
.610	80.35	1.45	82.5	65.03	1.44	41.25	55.59				
.541	103.64	1.58	105	78.29	1.68	52.5	68.17				
.491	125.90	1.70	127.5	91.18	1.94	63.75	80.78				
.453	148.47	1.82	150	104.52	2.15	75	93.15				
.422	170.99	1.96	172.5	118.17	2.42	86.25	106.16				
.397	192.59	2.02	195	131.68	2.66	97.5	117.98				
.376	214.25	2.09	217.5	144.36	2.88	108.75	130.79				

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REFERENCES

- CHOW, Y.S. and ROBBINS, H. (1965). On the asymptotic theory of fixed-width sequential confidence intervals for the mean. *Ann. Math. Statist.* **36**, 457-462.
- GEERTSEMA, J.C. (1968). Sequential confidence intervals based on rank tests. Ph.D. dissertation, University of California, Berkeley (unpublished).
- SIMONS, G. (1968). On the cost of not knowing the variance when making a fixed-width confidence interval for the mean. *Ann. Math. Statist.* **39**, 1946-1952.
- STARR, N. (1966). The performance of a sequential procedure for the fixed-width estimation of the mean. *Ann. Math. Statist.* **37**, 36-50.

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